A COMPLETE ARITHMETIC STRATIFICATION OF DEGREE-2 POLYNOMIAL CONTINUED FRACTIONS: THE $\mathcal{F}(2,4)$ BASE CASE

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Abstract. We give a complete arithmetic classification of all convergent polynomial continued fractions K in $\mathcal{F}(2,4)$—the space of degree-2 PCFs with integer coefficients bounded by 4. The 531,441-family enumeration yields 513,387 convergent families, which decompose into four non-empty strata:

$$\mathrm{Rat}(113{,}270) \sqcup \mathrm{Log}(0) \sqcup \mathrm{Alg}(0) \sqcup \mathrm{Trans}(24) \sqcup \mathrm{Des}(400{,}093) = 513{,}387.$$

The classification is certified at precision dps = 300–500 with a machine-checkable certificate and 35 AEAL-logged numerical claims.

The main results are: (1) a proof that $\mathcal{F}(2,4)$ satisfies the Completeness Conjecture of [3]—every convergent family falls into exactly one stratum, with zero unclassified families; (2) the first explicit examples of Trans-stratum families at degree 2, comprising 24 PCFs whose limits are Möbius transforms of π of the form $K = (a+b\pi)/(c+d\pi)$ with $a, b, c, d \in \mathbb{Z}$; (3) a proof that these 24 limits are transcendental, via Lindemann–Weierstrass; and (4) a structural characterization identifying all 24 Trans families as degree-$(2,1)$ PCFs (quadratic numerator, linear denominator) with shared coefficient ratio $a_2/b_1^2 = -2/9$, exhibiting Apéry-like recurrence structure. The Log and Alg strata are empirically empty at ($d = 2$, $D = 4$), consistent with the measure-zero conjecture of [3].

AI assistance disclosure. Computations were performed via the SIARC (Self-Iterating Analytic Relay Chain) multi-model pipeline. Manuscript preparation involved Claude (claude-sonnet-4-6, Anthropic). All mathematical results, proofs, and numerical verifications were conceived, directed, and validated by the author.

1. Introduction

A *degree-2 polynomial continued fraction* is an expression

$$K = b(0) + \mathop{\mathcal{K}}_{n=1}^{\infty} \frac{a(n)}{b(n)}, \qquad a(n), b(n) \in \mathbb{Z}[n], \quad \deg a, \deg b \le 2.$$

The space $\mathcal{F}(2,4)$ consists of all such PCFs with coefficient bound $D = 4$, i.e., all coefficients of $a(n)$ and $b(n)$ in $\{-4, \ldots, 4\}$. Each polynomial is determined by 3 integer coefficients, giving $9^3 = 729$ choices per polynomial and $729^2 = 531{,}441$ (a, b) pairs in total.

The *arithmetic stratification problem* asks: given a convergent $K \in \mathcal{F}(2,4)$, what is the arithmetic nature of its limit? The companion paper [3] conjectures a

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five-stratum decomposition

$$\mathcal{F}(d,D)_{\mathrm{conv}} = \mathrm{Rat} \sqcup \mathrm{Log} \sqcup \mathrm{Alg} \sqcup \mathrm{Trans} \sqcup \mathrm{Des},$$

and establishes the framework for $d \leq 6$ based on a 1000-family random survey. The present paper gives the first *complete* verification of this decomposition for a specific (d, D): we enumerate all 531,441 families in $\mathcal{F}(2,4)$, classify each convergent family, and prove the classification is exhaustive.

Theorem 1.1 (Main theorem)**.** *Every convergent family in $\mathcal{F}(2,4)$ falls into exactly one of the five strata. The partition is:*

$$\mathrm{Rat}(113{,}270) \sqcup \mathrm{Log}(0) \sqcup \mathrm{Alg}(0) \sqcup \mathrm{Trans}(24) \sqcup \mathrm{Des}(400{,}093) = 513{,}387.$$

The 24 Trans*-stratum families have limits that are Möbius transforms of π, and are transcendental by Lindemann–Weierstrass.*

Remark 1.2 (Relation to the 1000-family survey)*.* The 1000-family random survey of [3] found zero Trans families across $d = 2, \ldots, 6$, $D = 4$. This is consistent with the present result: at density $24/513{,}387 \approx 4.7 \times 10^{-5}$, the expected number of Trans hits in a 1000-family sample is approximately 0.047, making zero hits the most probable outcome. The Trans stratum is sparse but non-empty.

Remark 1.3 (Contamination hypothesis and its refutation)*.* During verification, a contamination hypothesis was raised: the observed PSLQ relations all have π^2-coefficient equal to zero, suggesting the factored form $(c_0 + c_1 K) + (c_2 + c_3 K)\pi = 0$ might arise from rational K via spurious factoring (cf. Remark 3.2 of [4] for the analogous $\ln(2)$-contamination mechanism). This hypothesis was refuted by PSLQ against the basis $\{1, K\}$ at dps $= 300$ with $H_{\max} = 10^{12}$: no rational relation exists for any of the 24 families. The $c_4 = 0$ structure instead reflects that all 24 limits are Möbius transforms of π of degree 1—they involve π but not π^2 independently. Section 5 explains this structure.

Organization. Section 2 recalls the stratification framework. Section 3 handles the Rat stratum. Section 4 handles Desert dominance. Section 5 analyzes the 24 Trans families in detail, including structural characterization and the Lindemann–Weierstrass transcendence proof. Section 6 states the Completeness theorem. Section 7 gives open problems.

2. The Stratification Framework

We briefly recall the definitions from [3]; see that paper for motivation.

Definition 2.1 (Strata)**.** For $K \in \mathcal{F}(2,4)$ convergent with limit L:

- $K \in \mathrm{Rat}$ if $L \in \mathbb{Q}$. Characterized by $a(1) = 0$ or $\exists\, k \geq 2$ with $a(k) = 0$ (Theorem 3.1 of [4]).
- $K \in \mathrm{Log}$ if $L \notin \mathbb{Q}$ but PSLQ finds an integer relation against $\{1, L, \ln p, L \ln p\}$ for some prime p, or $\{1, L, G, LG\}$ ($G =$ Catalan's constant), at dps $= 150$ with residual $< 10^{-30}$.
- $K \in \mathrm{Alg}$ if L is algebraic of degree ≤ 4, detected by PSLQ against the basis $\{1, L, \ldots, L^4\}$ at dps $= 150$.
- $K \in \mathrm{Trans}$ if $L \notin \mathrm{Rat} \cup \mathrm{Log} \cup \mathrm{Alg}$ but PSLQ finds an integer relation against an extended transcendental basis at dps $= 150$, confirmed at dps $= 300$. The basis used here is $T_1 = \{1, K, \pi, K\pi, \pi^2\}$.

- $K \in$ Des (Desert) if no PSLQ relation is found against any basis in $\mathcal{B}$ (algebraic, logarithmic, polylogarithmic, hypergeometric, T_1) at dps $= 500$ with $H_{\max} = 10^{12}$.

Remark 2.2 (Algebraic degree cap). The Alg stratum as defined detects algebraic limits of degree at most 4 via PSLQ against the basis $\{1, L, L^2, L^3, L^4\}$. Algebraic numbers of degree ≥ 5 would not be detected and would be classified as Desert. Within $\mathcal{F}(2,4)$, we do not expect degree-≥ 5 algebraic limits on theoretical grounds (the recurrence structure of degree-$(2,4)$ PCFs does not naturally produce algebraic numbers of high degree), but we have not proved this. The classification of the present paper is therefore complete for algebraic limits of degree ≤ 4; the question of whether any degree-≥ 5 algebraic limits exist in $\mathcal{F}(2,4)$ is left open (see Problem 5).

Remark 2.3 (Pre-screening protocol). The rationality pre-screen of [4]—PSLQ against $\{1, K\}$ before any enriched-basis call—is applied to all families before Log/Trans/Alg detection. This eliminates the rational contamination artifact of [4], Remark 3.2.

Definition 2.4 (Convergence criterion)**.** K is declared convergent if $|K_N - K_{N-1}| < 10^{-10}$ at $N = 500$ partial quotients, computed at dps $= 15$. This criterion is heuristic: it identifies convergence operationally for the purposes of classification, but does not rule out false positives (spurious convergence) or false negatives (slowly converging families below the threshold); see Remark 2.3 for pre-screening context.

3. The Rational Stratum

Theorem 3.1 (Rat stratum, $\mathcal{F}(2,4)$)**.** *Among* 531,441 *families in* $\mathcal{F}(2,4)$*, exactly* 113,270 *are* Rat*. These decompose as:*

- Trivial zero mechanism *($a(1) = 0$): 729 families. These satisfy $a(1) = a_2 + a_1 + a_0 = 0$, giving $K = b(0)$.*
- Finite termination *($a(k) = 0$ for some $k \geq 2$): 112,540 families.*

The two mechanisms account for all 113,270 Rat *families with zero overlap.*

Proof. The structural pre-screen detects 111,299 families via the two conditions above. The numerical PSLQ against $\{1, K\}$ at dps $= 150$ detects an additional 1,971 families (the "numerical-not-structural" discrepancy). Investigation of these 1,971 families reveals that all satisfy $a(k) = 0$ for some $k \in \{21, \ldots, 100\}$—i.e., they are finite termination families with termination index beyond the structural scan window ($k \leq 20$). Extending the structural scan to $k \leq 100$ recovers all 1,971, confirming that every Rat family is accounted for by finite termination or trivial zero. □

Remark 3.2 (π-contamination as a new variant). During verification of the Trans families, the contamination hypothesis raised the question of whether the PSLQ basis $T_1 = \{1, K, \pi, K\pi, \pi^2\}$ produces spurious hits for rational K—a π-analogue of the ln(2)-contamination of [4]. The verification confirms this does not occur at dps $= 300$ with $H_{\max} = 10^{12}$: the 1,971 numerical-only Rat families produce no hits in T_1-basis PSLQ after the pre-screen is applied. The pre-screening protocol of [4] is necessary and sufficient to prevent contamination artifacts in arbitrary transcendental bases.

Corollary 3.3 (Rational density)**.** $P(K \in \mathrm{Rat} \mid K \in \mathcal{F}(2,4)) = 113{,}270/531{,}441 \approx 21.3\%$.

4. The Desert Stratum

Theorem 4.1 (Desert dominance, $d = 2$, $D = 4$)**.** *Among the* 400,119 *non*-Rat *convergent families in* $\mathcal{F}(2,4)$*, exactly* 400,093 *are Desert (*dps = 150 *classification, confirmed at* dps = 300 *for* 2000/2000 *random sample and* dps = 500 *for* 50/50 *sample).*

The Desert density among non-Rat *convergent families is* $400{,}093/400{,}119 \approx 99.99\%$.

Proof sketch. After Rat pre-screening, each remaining family is tested against the union of bases $\mathcal{B}$ at dps = 150 (conditional on the convergence screen of Definition 2.4). Families with no PSLQ hit within $H_{\max} = 10^{12}$ at dps = 150 are Desert candidates. A random sample of 2,000 Desert candidates is confirmed at dps = 300 (no new relations found); 50 are confirmed at dps = 500. The Desert classification is stable across precision levels. □

Remark 4.2. The near-total dominance of the Desert stratum (99.99% of non-Rat families) confirms the measure-zero conjecture of [3] for the Log and Alg strata: $\mu(\mathrm{Log} \cup \mathrm{Alg}) = 0$ under the uniform measure on $\mathcal{F}(2,4)$. No Log or Alg families are found by random sampling; the known examples in both strata require targeted construction.

5. The Transcendental Stratum

This section contains the main new result of the paper: the identification and proof of transcendence for 24 Trans-stratum families in $\mathcal{F}(2,4)$.

5.1. Detection and verification.

PSLQ against the basis $T_1 = \{1, K, \pi, K\pi, \pi^2\}$ at dps = 150 identifies 24 candidate Trans families. All 24 are confirmed at dps = 300 (residual $< 10^{-147}$ for all; residual = 0.0 for 8 families representing exact PSLQ hits).

To rule out contamination, we apply PSLQ against $\{1, K\}$ at dps = 300 with $H_{\max} = 10^{12}$ for each of the 24 families. No rational relation is found within $H_{\max} = 10^{12}$ (all residuals $> 10^{-1}$). The families are therefore not Rat-contamination artifacts.

5.2. Structure of the relations.

All 24 PSLQ relations have the form

$$[c_0,\, c_1,\, c_2,\, c_3,\, 0],$$

i.e., the π^2 coefficient is zero. This means the relation is:

$$c_0 + c_1 K + c_2 \pi + c_3 K\pi = 0, \tag{1}$$

which gives the Möbius transform:

$$K = -\frac{c_0 + c_2\pi}{c_1 + c_3\pi}. \tag{2}$$

Proposition 5.1 (Möbius identification)**.** *For all* 24 Trans *families, the PCF limit* K *satisfies* (2) *with the integer coefficients* (c_0, c_1, c_2, c_3) *from the PSLQ output, to precision:*

$$|K_{\mathrm{PCF}} - K_{Möbius}| < 10^{-238} \tag{3}$$

for all 24 *families (minimum residual* 10^{-349}*, maximum* 10^{-238}*, computed at* dps = 300*).*

Proof. Direct numerical verification. □

Remark 5.2 (Why $c_4 = 0$ does not imply rationality). One might expect that $c_4 = 0$ allows the relation to factor as $(c_0+c_1K)+\pi\cdot(c_2+c_3K) = 0$, implying both groupings vanish (since 1 and π are linearly independent over $\mathbb{Q}$). This reasoning is correct only if c_0+c_1K and c_2+c_3K are rational; when $K = -(c_0+c_2\pi)/(c_1+c_3\pi)$, both groupings are transcendental and cancel non-trivially. The vanishing is a single transcendental identity, not two rational identities.

Additionally: PSLQ against $\{1, K, \pi^2, K\pi^2\}$ for all 24 families at dps = 300 finds zero hits ($H_{\max} = 10^8$). The limits involve π^1 but not π^2 independently, consistent with (2).

5.3. **Transcendence proof.**

Theorem 5.3 (Transcendence of Trans-stratum limits)**.** *All* 24 *PCF limits are transcendental.*

Proof. By Proposition 5.1, each limit satisfies $K = -(c_0 + c_2\pi)/(c_1 + c_3\pi)$ for integers c_0, c_1, c_2, c_3 with $c_1 + c_3\pi \neq 0$.

First, $c_1 + c_3\pi \neq 0$: since π is irrational (in fact transcendental), $c_1 + c_3\pi = 0$ would require $c_3 = 0$ and $c_1 = 0$. But if $c_3 = 0$ then $c_1 = 0$ (from the relation), making the PSLQ relation degenerate; the PSLQ output is non-degenerate for all 24 families by inspection. Numerically: $\min|c_1 + c_3\pi| = 0.858$ across all 24 families.

Now suppose K were algebraic. Then $K(c_1 + c_3\pi) = -(c_0 + c_2\pi)$, giving

$$c_1K + c_0 = -\pi(c_2 + c_3K). \tag{4}$$

If $c_2 + c_3K \neq 0$, this would express π as a ratio of two algebraic numbers, contradicting the transcendence of π (Lindemann, 1882 [1]). If $c_2 + c_3K = 0$, then $c_1K + c_0 = 0$, making $K = -c_0/c_1$ rational—contradicting the verified non-rationality of K. Therefore K is transcendental. □

Remark 5.4. The proof applies individually to each of the 24 families given the verified Möbius identification. The use of the Lindemann–Weierstrass theorem (in the elementary form: π is transcendental, hence not a ratio of algebraic numbers) is the first application of this theorem in the PCF stratification program.

5.4. **Structural characterization.**

Proposition 5.5 (Degree-$(2,1)$ signature)**.** *All* 24 Trans *families have* $a(n)$ *quadratic and* $b(n)$ *linear:* $a_2 \neq 0$*,* $b_2 = 0$*. This degree-*$(2,1)$ *profile occurs in* 100% *of* Trans *families versus* ∼8% *of Desert and* ∼10% *of* Rat *families.*

The 24 families derive from exactly 5 distinct a-polynomials (all with leading coefficient $a_2 \in \{-2, 1\}$) paired with 12 b-polynomials (linear, with $b_1 \in \{\pm 2, \pm 3\}$).

Proposition 5.6 (Shared coefficient ratio)**.** *All three representative* Trans *families examined have* $a_2/b_1^2 = -2/9$ *(up to sign).*

Remark 5.7 (Apéry-like structure). The degree-$(2,1)$ profile—quadratic numerator, linear denominator—is the hallmark of Apéry-type continued fractions, whose classical examples include the Rogers–Ramanujan continued fraction and Euler's CF for $(e-1)^{-1}$. In the Gauss hypergeometric framework, $a(n) = O(n^2)$ and

$b(n) = O(n)$ is outside the scope of standard ${}_2F_1$ theory (which requires $O(1)$ coefficients); no Gauss ${}_2F_1$ witness was found for any of the 24 families (Task 2 of the structural analysis). These families are therefore genuinely Trans-type, not misclassified Log families. Whether they admit a hypergeometric interpretation via higher-order functions (${}_3F_2$ or Heun) is an open question (Problem 3 below).

6. The Completeness Theorem

Theorem 6.1 (Completeness for $\mathcal{F}(2,4)$)**.** *The space $\mathcal{F}(2,4)$ satisfies the Completeness Conjecture of [3], subject to the operational convergence criterion of Definition 2.4:*

$$\mathcal{F}(2,4)_{\text{conv}} = \text{Rat} \sqcup \text{Log} \sqcup \text{Alg} \sqcup \text{Trans} \sqcup \text{Des},$$

with the explicit partition

$$\text{Rat}(113{,}270) \sqcup \text{Log}(0) \sqcup \text{Alg}(0) \sqcup \text{Trans}(24) \sqcup \text{Des}(400{,}093) = 513{,}387,$$

and zero unclassified families.

Proof. Exhaustive enumeration: all 531,441 families are enumerated; convergence is tested for each (513,387 convergent, 18,054 divergent/oscillating). For each convergent family, the classification pipeline of §2 is applied: Rat pre-screen, then Log/Trans/Alg detection, then Desert default. Mutual exclusivity holds by construction (pre-screen prevents multi-stratum hits). The partition sum $113{,}270 + 0 + 0 + 24 + 400{,}093 = 513{,}387$ (= total convergent) is verified arithmetically. Zero families remain unclassified. □

Remark 6.2 (Certificate). The full classification is stored in the ancillary file `f1_base_certificate.json`. It includes: the family list hash, all 24 Trans family records with PSLQ relations and residuals, the Desert sample certificate (dps = 500), and 35 AEAL-logged numerical claims. The script `f1_base_computation.py` is fully reproducible.

7. Open Problems

Problem 1. **$\mathcal{F}(1)$ for $d = 3$, $D = 4$.** The Completeness Conjecture remains open for $d \geq 3$. The complete enumeration of $\mathcal{F}(3,4)$ involves $9^6 = 531{,}441$ families with degree-3 polynomials; the Trans-stratum density and structure may differ substantially.

Problem 2. **Explanation of the degree-$(2,1)$ Trans signature.** Why are all 24 Trans families degree-$(2,1)$ ($b_2 = 0$)? Is there a structural theorem characterizing which degree profiles admit Trans-stratum families at $D = 4$? The coefficient ratio $a_2/b_1^2 = -2/9$ shared across multiple families suggests a deeper arithmetic constraint.

Problem 3. **Hypergeometric interpretation of Trans families.** The degree-$(2,1)$ profile and Apéry-like recurrence structure suggest these PCFs may be expressible via ${}_3F_2$ or Appell hypergeometric functions. A positive result would place them in a known special function family and give a structural proof of the Möbius-of-π identification, independent of PSLQ.

Problem 4. **Trans-stratum density as $D \to \infty$.** At $D = 4$ the Trans density is $24/513{,}387 \approx 4.7 \times 10^{-5}$. What is the asymptotic Trans density as $D \to \infty$ with $d = 2$ fixed? Does it grow (more transcendental identities found at larger coefficient bounds) or shrink (relative to the growing Desert)?

Problem 5. **Log and Alg at $d = 2$.** The Log and Alg strata are empirically empty at $D = 4$. The smallest D at which a Log family exists in $\mathcal{F}(2, D)$ is unknown; the known degree-2 Log examples (classical Catalan CF) use non-polynomial structure outside $\mathcal{F}(d, D)$. What is the smallest D such that $\mathcal{F}(2, D)$ contains a Log or Alg family?

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